

Monroe County Heat-Day Trend Analysis Using a Heat Stress Framework (1994–2025)

Executive Summary

June is now the key early-season heat-risk month, with repeated multi-day heat events, earlier onset, and a clear need for cooling-center readiness by June.

1 Heat-Day Trend Analysis

Heatwave days were defined using a combined temperature and relative humidity risk framework. Heat stress risk rises when high temperatures occur alongside elevated humidity because moisture reduces evaporative cooling from sweating, which increases the perceived temperature through the Heat Index. To keep the analysis consistent with the available daily temperature record, this report flags days with daily maximum temperature $\geq 88^{\circ}\text{F}$ as heatwave days, especially when consecutive hot days occur.

1.1 Heat Risk Definition

The analysis follows NOAA-style Heat Index logic, where higher heat index values imply greater health risk for exposed populations. As a practical screening rule, days with temperatures around $88\text{--}92^{\circ}\text{F}$ become increasingly concerning when relative humidity is moderate to high, because the apparent temperature can move into “Extreme Caution” or “Danger” conditions. This approach is appropriate for public-health planning because it captures environmental conditions that can trigger heat illness, especially for older adults, outdoor workers, and people without reliable cooling.

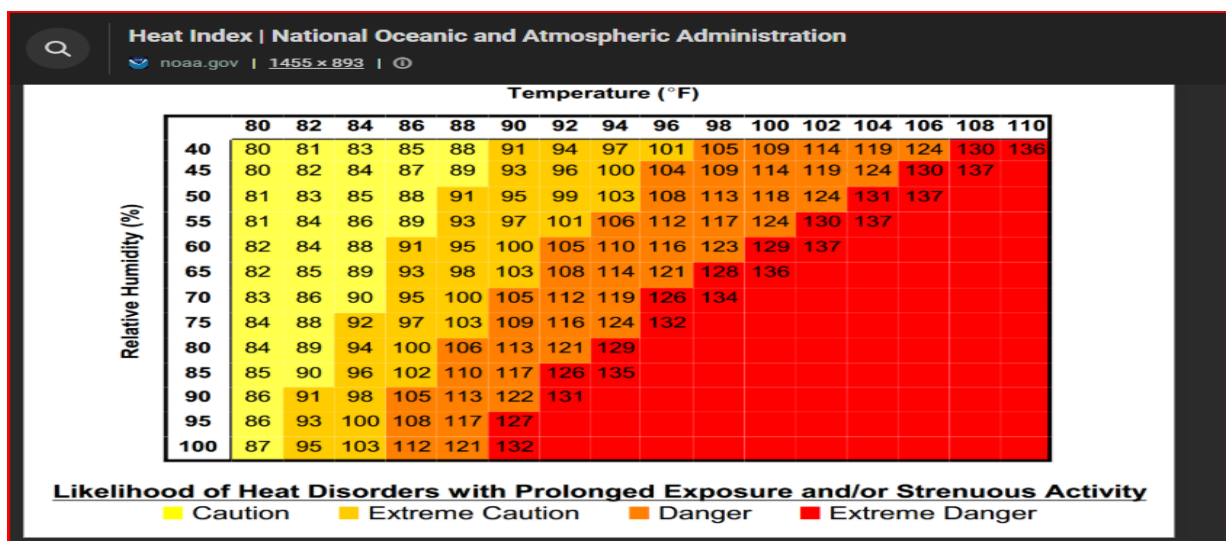


Figure 1: Temperature and humidity relationship used to frame heat stress risk

2 Analysis Summary Table of Historical Data

Table 1: Combined monthly heat-day summary, 1994–2025

Month	Avg. Heat Days	Max Heat Days	Median Heat Days	Heat Years
May	1.40	3	1	15
June	2.93	6	3	27
July	6.00	14	6	26
August	4.56	12	4	18

Insights

July remains the peak month for heat-day frequency, but June is now an important early-season heat month, appearing in 27 years compared with 26 years for July. August is also substantial, which shows that high heat extends well beyond mid-summer.

2.1 Early-Season Heat

June now shows repeated multi-day heat events and an earlier seasonal onset, including a June 4 start in 2025. This is the strongest evidence that cooling-center readiness should shift earlier than July.

Table 2: May and June heat days with timing details

Month	Year	Days $\geq$ 88F	First Heat Day	Last Heat Day	Mean Heat Temp	Timeline Observed
May	2007	3	24	31	88.7	24, 25, 31
May	2006	2	29	30	91.5	29, 30
May	2013	2	30	31	88.5	30, 31
May	2015	2	8	9	90.5	8, 9
May	2016	2	27	28	89.5	27, 28
May	1996	1	19	19	89.0	19
May	1998	1	16	16	88.0	16
May	2010	1	31	31	88.0	31
May	2011	1	31	31	91.0	31
May	2012	1	28	28	91.0	28
May	2018	1	30	30	92.0	30
May	2020	1	26	26	92.0	26
May	2021	1	21	21	88.0	21
May	2022	1	31	31	88.0	31
May	2024	1	21	21	88.0	21
June	2021	6	6	29	90.5	6, 7, 26, 27, 28, 29
June	2012	5	11	30	89.8	11, 20, 21, 29, 30
June	1994	4	15	18	93.0	15, 16, 17, 18
June	1999	4	6	27	89.8	6, 7, 12, 27
June	2001	4	14	29	90.8	14, 15, 19, 29
June	2005	4	9	27	90.0	9, 24, 25, 27
June	2007	4	8	27	90.8	8, 12, 26, 27
June	2008	4	6	13	90.8	6, 8, 9, 13
June	2016	4	19	27	91.0	19, 20, 26, 27
June	2018	4	17	30	89.8	17, 18, 29, 30
June	2022	4	16	30	90.5	16, 21, 22, 30
June	2024	4	17	20	91.8	17, 18, 19, 20
June	2025	4	4	24	90.2	4, 22, 23, 24

2.2 Mid- and Late-Summer Heat

July remains the most intense month, but June is the more operationally important early warning month because it is now showing consistent multi-day events. August also contributes major heat years, which means the summer heat risk window is broad rather than sharply concentrated in one month.

Table 3: July and August heat days with timing details

Month	Year	Days $\geq$ 88F	First Heat Day	Last Heat Day	Mean Heat Temp	Timeline Observed
July	1999	14	3	31	89.5	3, 4, 5, 6, 15, 16, 17, 18, 23, 24, 27, 28, 30, 31
July	2018	12	1	20	91.2	1, 2, 3, 4, 5, 9, 10, 13, 14, 15, 16, 20
July	2002	10	1	31	89.8	1, 2, 4, 8, 17, 21, 22, 28, 29, 31
July	2011	10	10	23	91.4	10, 11, 12, 16, 17, 18, 20, 21, 22, 23
July	2016	10	5	27	90.6	5, 6, 7, 12, 13, 14, 21, 22, 23, 27
July	2012	9	4	31	91.2	4, 6, 12, 13, 14, 16, 17, 23, 31
July	2005	8	4	26	89.9	4, 11, 12, 13, 14, 18, 25, 26
July	2020	8	2	19	91.0	2, 5, 6, 7, 8, 9, 10, 19
July	2025	8	5	24	90.1	5, 6, 7, 11, 12, 13, 15, 24
August	2016	12	3	28	91.0	3, 4, 5, 9, 10, 11, 12, 13, 16, 20, 24, 28
August	2002	9	1	16	91.0	1, 2, 4, 11, 12, 13, 14, 15, 16
August	1995	8	1	31	90.0	1, 3, 14, 15, 19, 20, 21, 31
August	2001	7	1	9	91.3	1, 2, 5, 6, 7, 8, 9
August	2007	6	1	29	90.8	1, 2, 3, 24, 25, 29
August	2018	5	4	29	91.2	4, 5, 6, 28, 29
August	2021	5	9	28	88.6	9, 11, 25, 26, 28
August	2022	5	3	29	89.2	3, 6, 8, 20, 29

3 EDA Figures

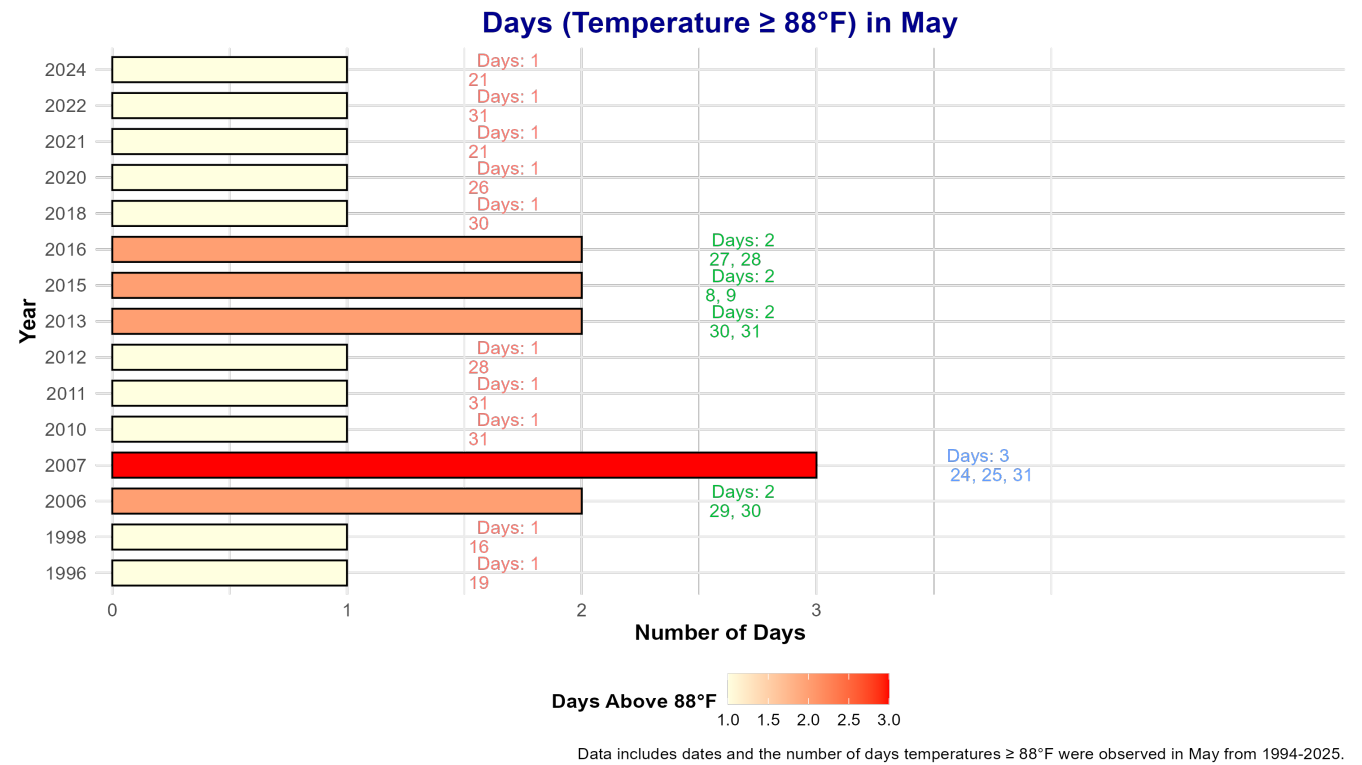


Figure 2: May heatwave days by year, 1994–2025

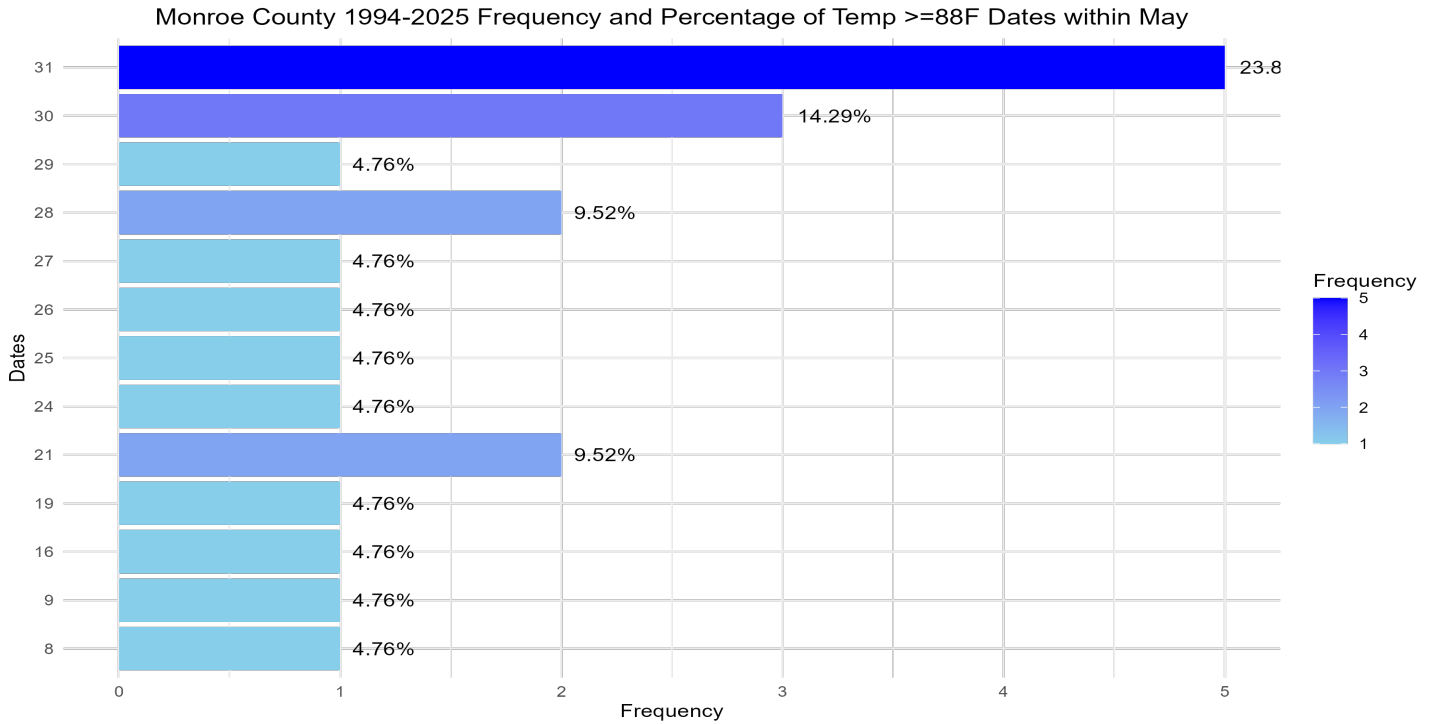


Figure 3: May heatwave frequency trends, 1994–2025

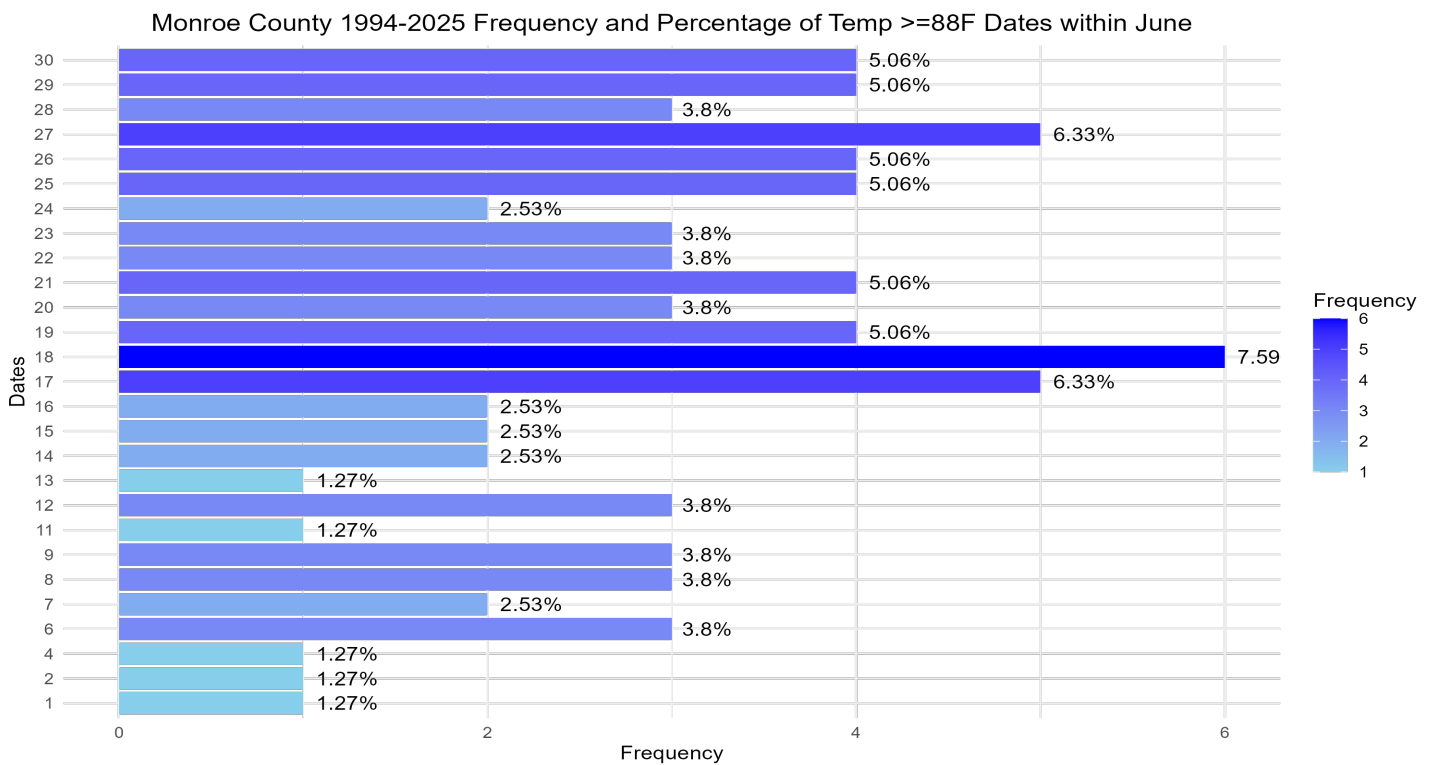


Figure 4: June heatwave frequency trends, 1994–2025

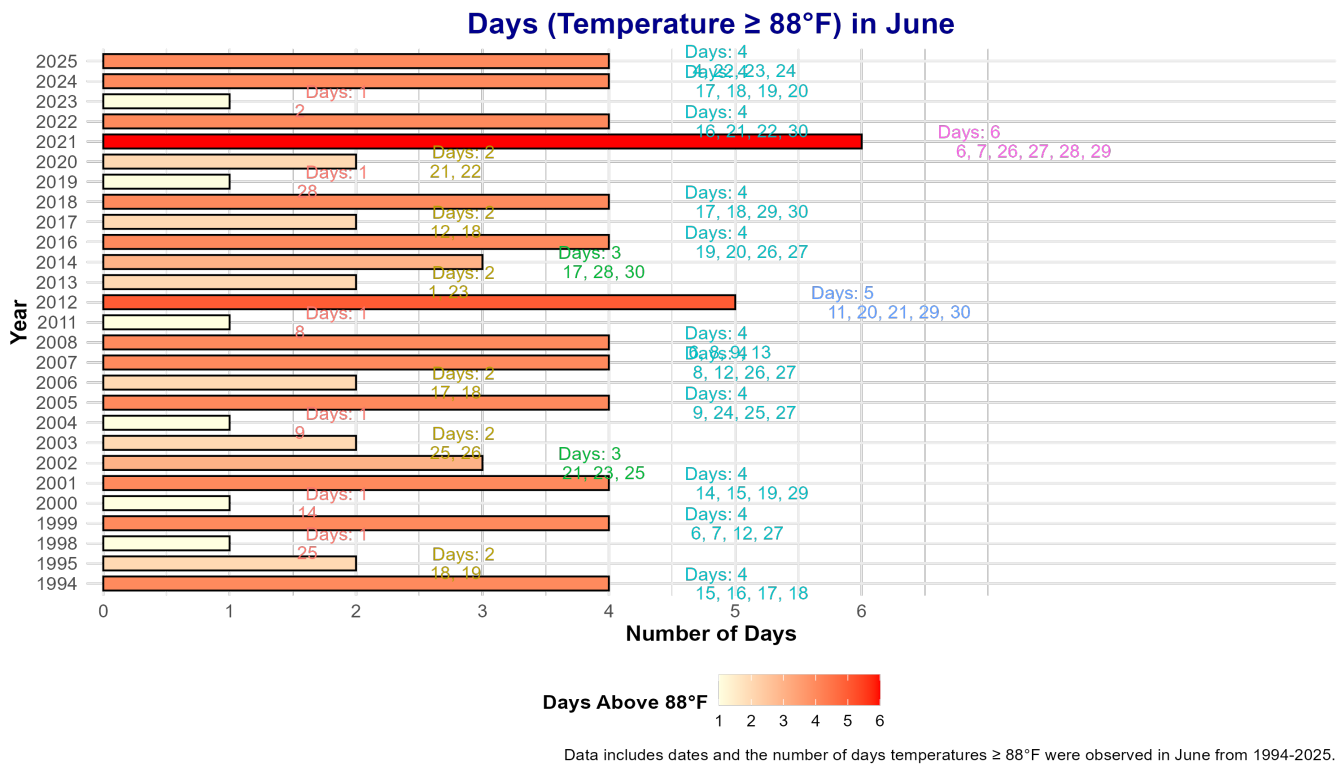


Figure 5: June heatwave days by year, 1994–2025

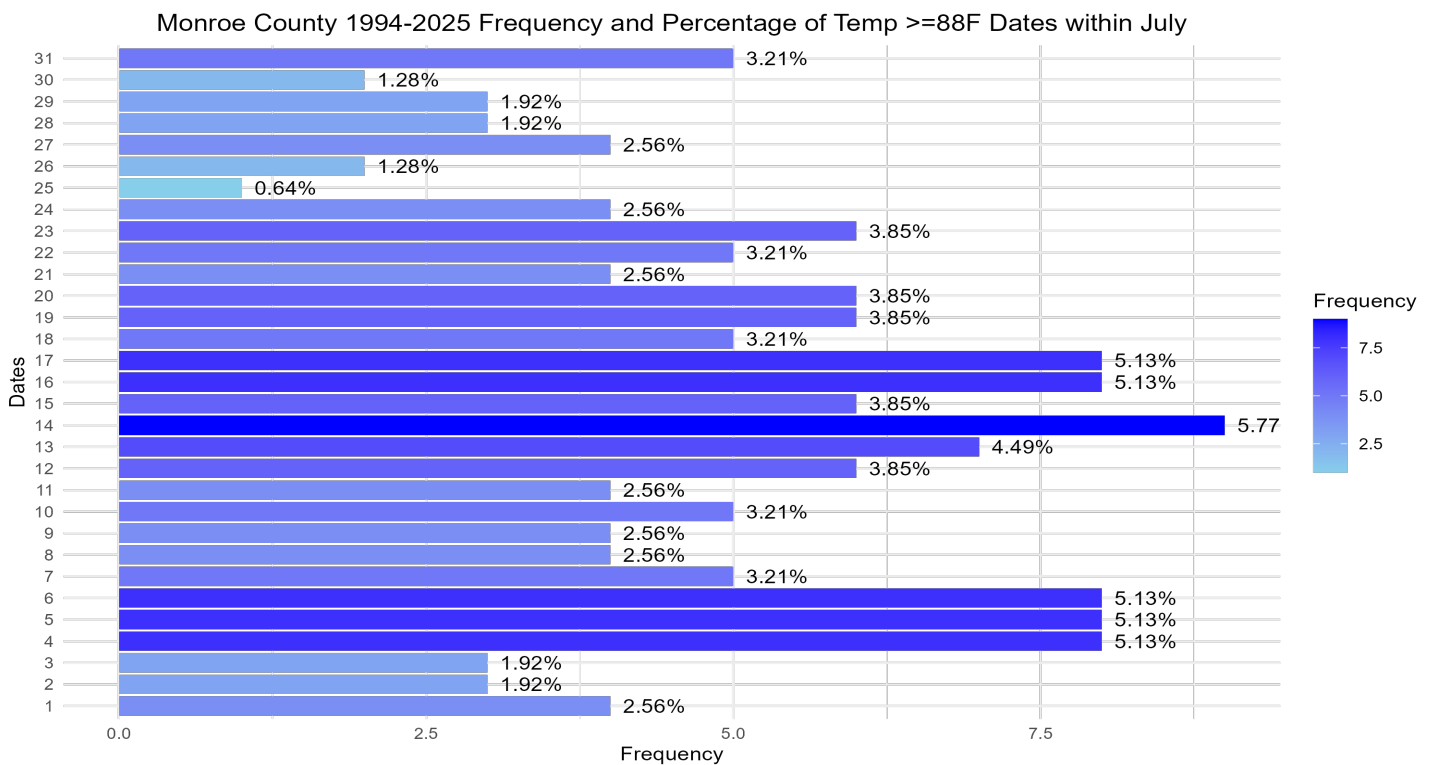


Figure 6: July heatwave frequency trends, 1994–2025

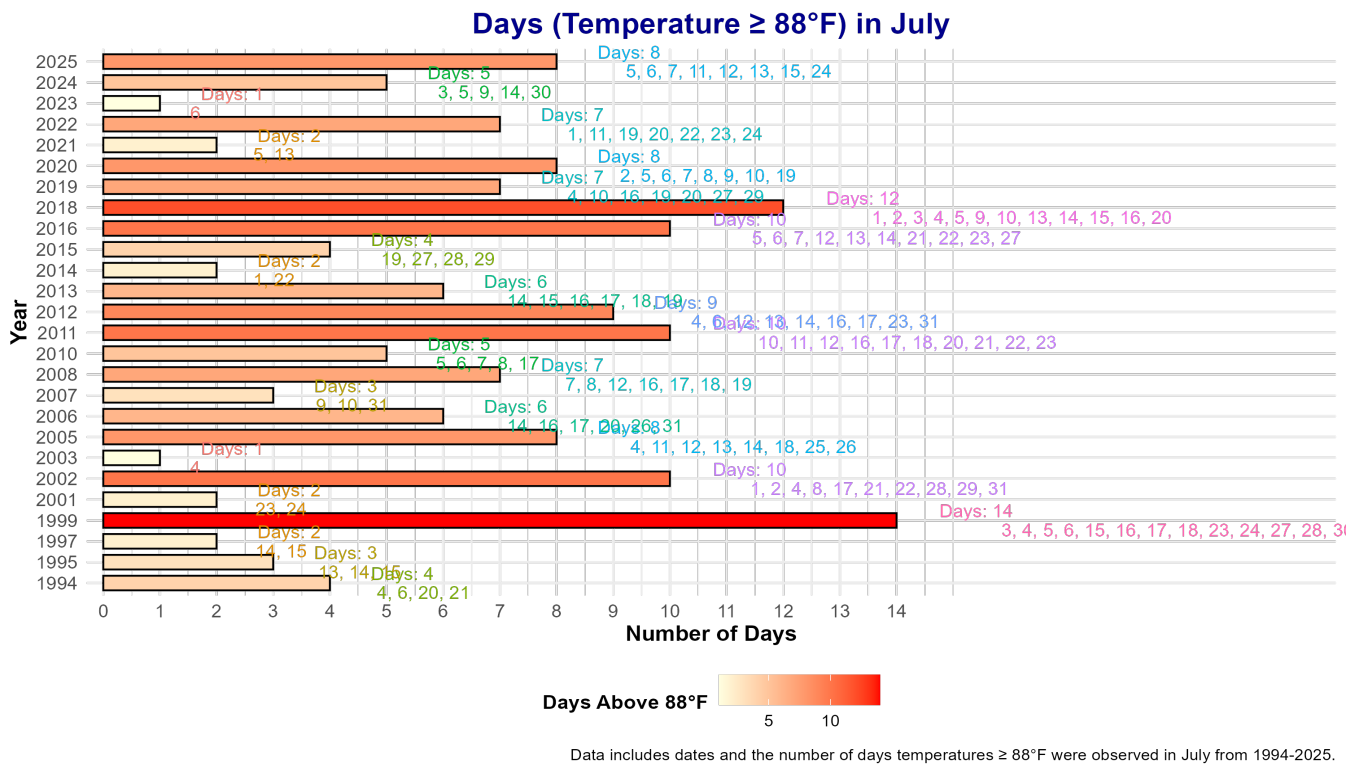


Figure 7: July heatwave days by year, 1994–2025

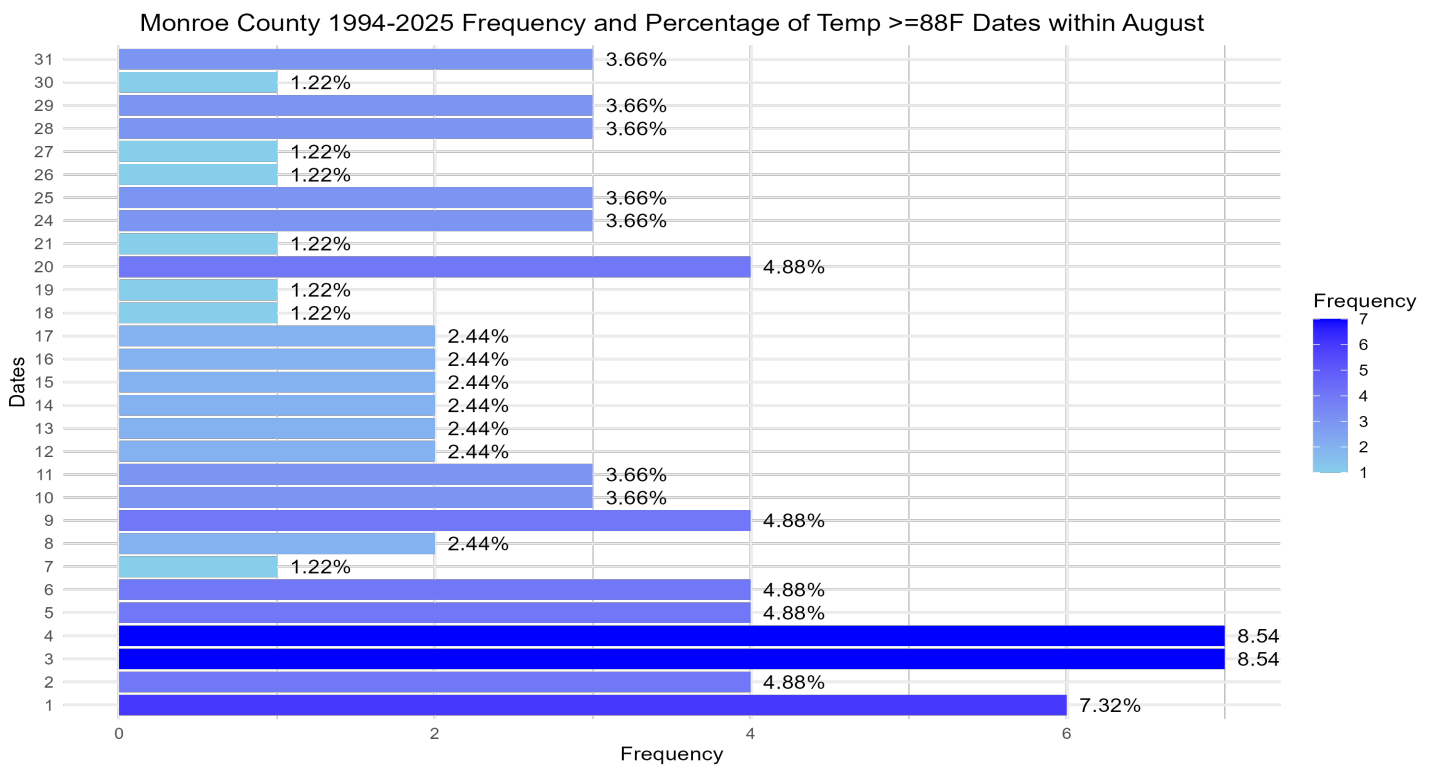


Figure 8: August heatwave frequency trends, 1994–2025

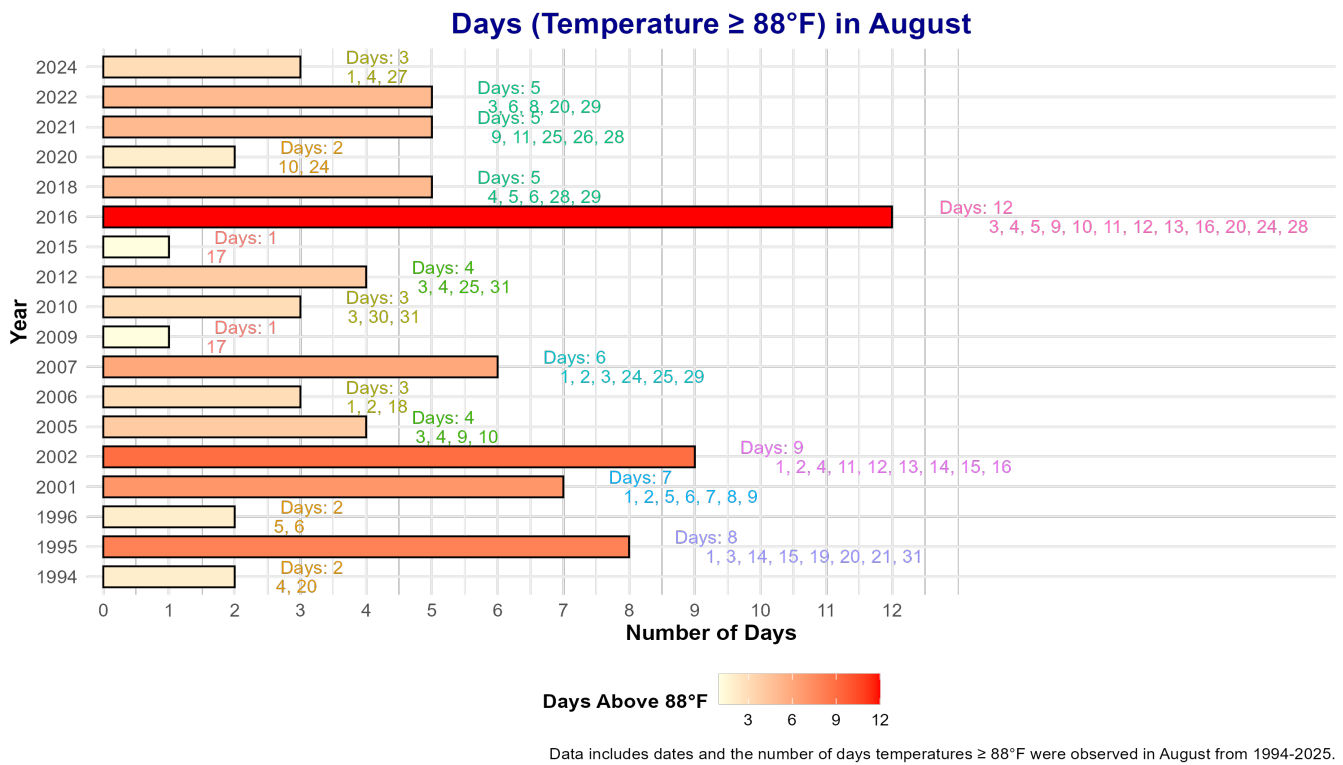


Figure 9: August heatwave days by year, 1994–2025

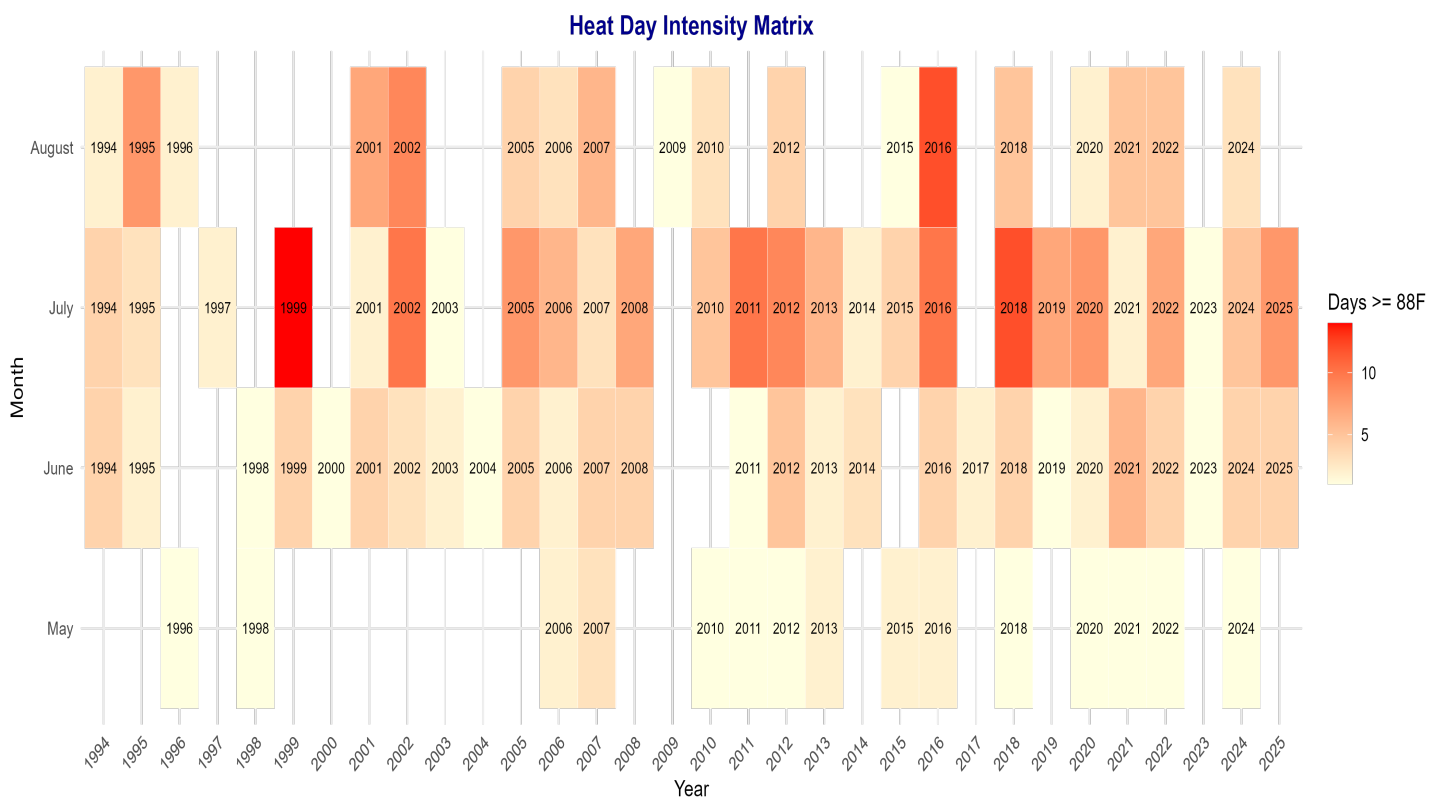


Figure 10: Heatwave intensity heatmap by month and year

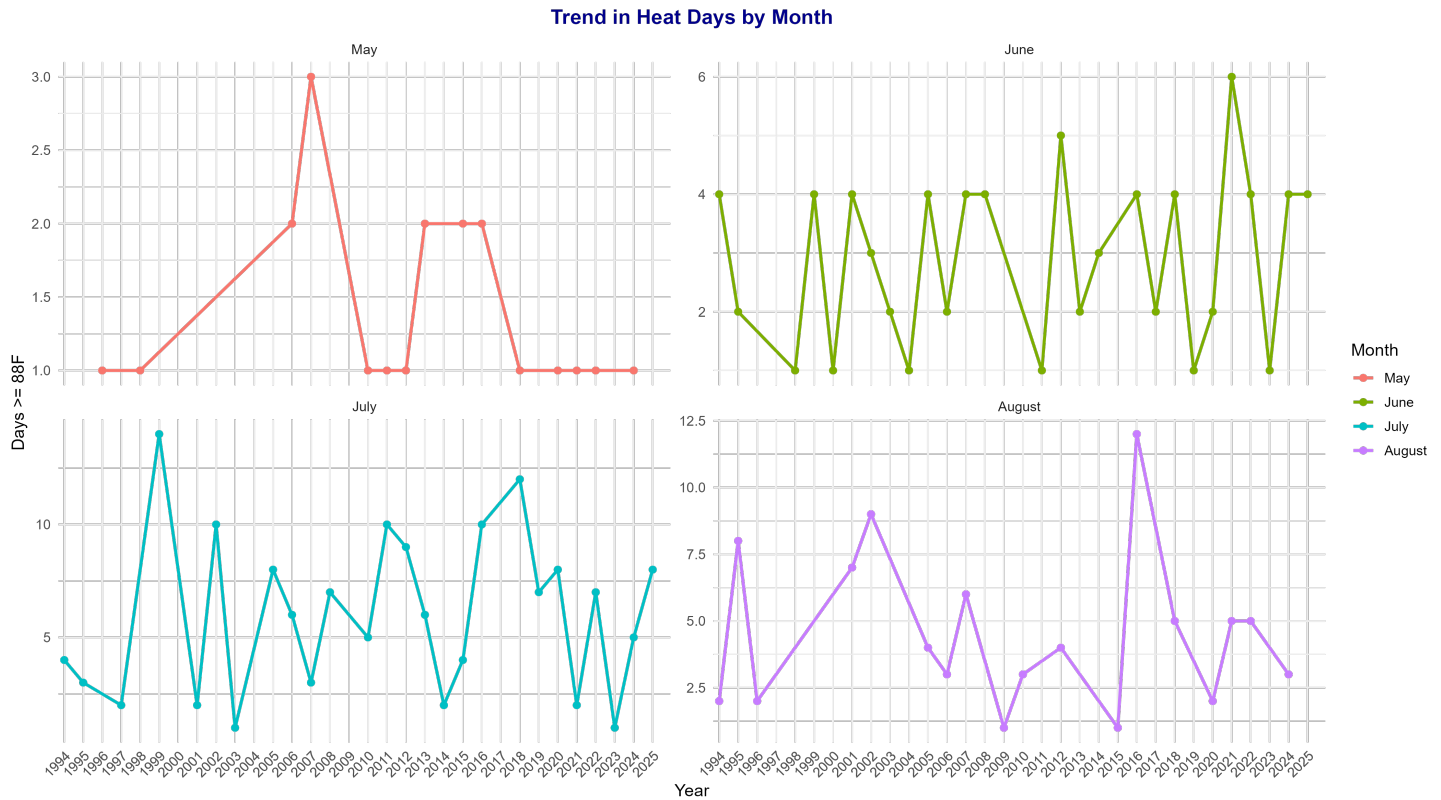


Figure 11: Trend in heat days by month, 1994–2025

4 Public Health Interpretation

May shows rare but real early-season heat, with the strongest activity clustered in late May and occasional single-day spikes earlier in the month. June is the clearest emerging risk month because it contains repeated multi-day events, earlier first heat dates, and the highest number of active heat years among the early-season months. July remains the peak month overall, but it is no longer the only month that matters for planning, and August still contributes important heat years, which shows that the risk window extends beyond midsummer.

4.1 High-Risk Time Windows

The table evidence supports four practical watch periods. May 21–31 is the emerging early-season window because the strongest May heat days cluster late in the month. June 4–30 should be treated as the main early-season alert period because June has the earliest consistent heat onset and repeated multi-day streaks, including the June 4 start in 2025. Mid-July remains the peak-risk window for the highest and longest heat bursts, while early to late August remains a secondary but still important heat-risk period.

4.2 Operational Recommendations

Key Takeaway

July remains the peak month for heat-day frequency, but June is now an important early-season heat month and should be treated as a public-health trigger period.

- Move to active cooling-center messaging when forecasts show sustained 88°F conditions or open cooling-center preparedness in early June.
- Use 2 or more consecutive days at or above 88 F in June as a practical trigger for staffing and outreach.
- Keep July as the peak-response month, but do not wait until July to begin readiness.
- Maintain an August watch status, especially when early August heat clusters appear.

4.3 Conclusion

The updated 1994–2025 record shows that dangerous heat is not confined to July. June has become a meaningful heat month with repeated multi-day events and earlier seasonal onset, and May contains occasional early-season heat that should not be ignored. The strongest policy signal is that heat is now arriving earlier, especially in June, so cooling-center planning should begin earlier than July, with readiness in early June and active communication by mid-June in years with sustained hot conditions.